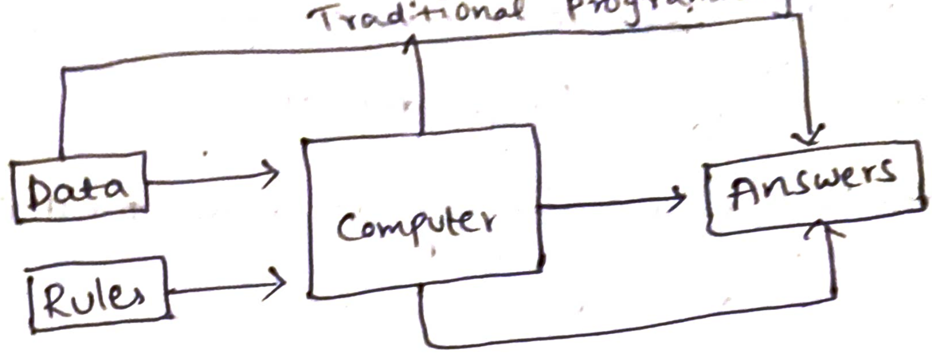
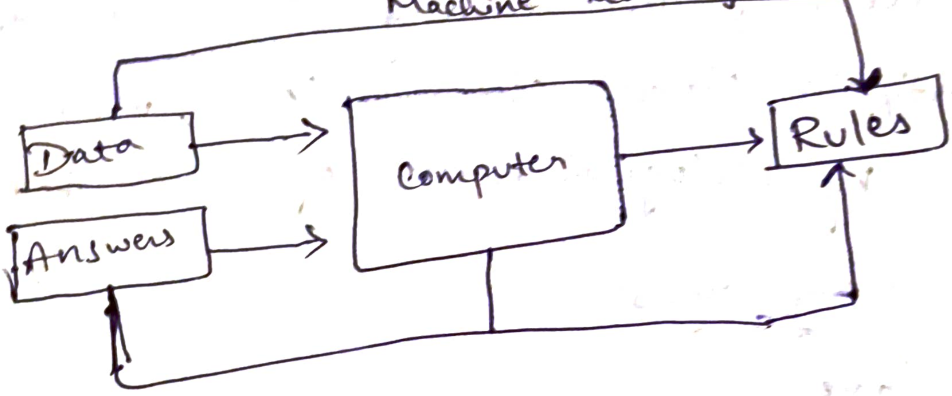
Deep learning

Computers write Their Own Rules

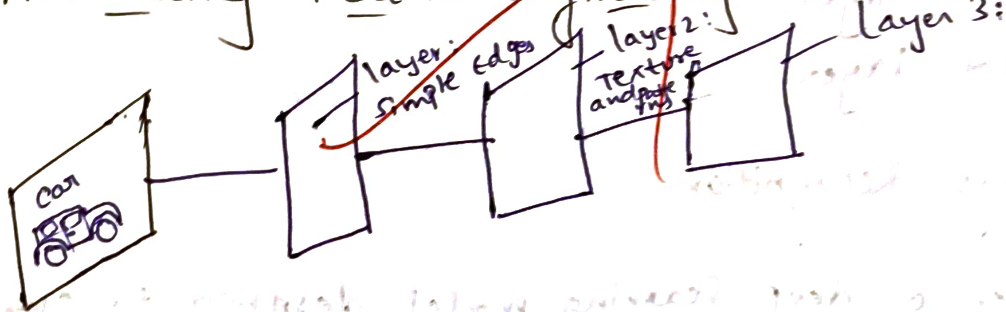
Traditional programming



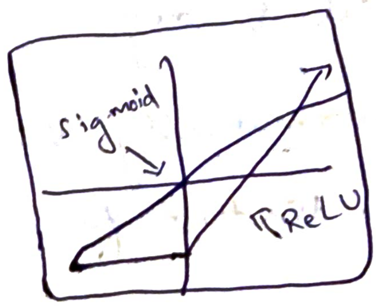
Machine learning - 1990s



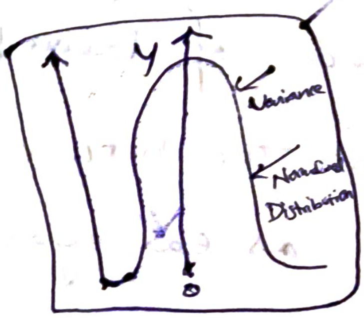
Automating feature engineering



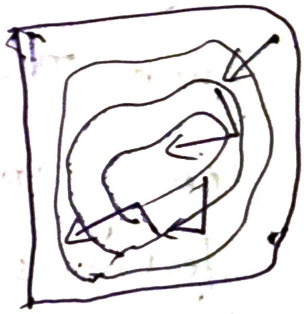
Algorithms :- The spark that enabled Depth



Better Activation Functions (ReLU)



Improved weight - Initialization Schemes



Sophisticated optimization (Adam)

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* Deep learning is a subset of machine learning that uses multi-layered artificial neural networks to automatically ~~can~~ learn complex patterns and representation from large datasets.

* Deep learning enable highly accurate predictions, classification, and decision-making.

- The foundation of deep learning lies in the structure of the Artificial Neural Network (AN), which is inspired by biological neural of human brain

Three primary layers:

→ Input layer.

→ Hidden layer

→ output layer

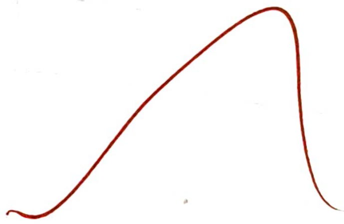


Image Recognition

Consider a deep learning model designed to classify images of cats and dogs.

- Input: Thousands of labeled cat & dog images

Learning Process: The network automatically learns features such as ear, eyes, fur texture, facial shapes etc.,

Output: Correctly classifies a new image as either a cat or a dog.

ML	DL
• Requires manual feature engineering	• Learns features automatically
• works well with small to medium datasets	• performs best with large datasets
• Uses simpler algorithms	• Uses deep neural networks with multiple hidden layers
• Training is comparatively faster	• Training is computationally intensive
• suitable for structured data	• Excels with structured and unstructured data

Learning Algorithm :

• Learning Algorithms are the mathematical techniques that enables a deep learning model to learn patterns from data by minimum prediction ~~eggs~~ errors

The Flywheel of Learning

Experience (E) → Task (T) → Performance (P) → Experience (E)

Three Essential components

$\boxed{\text{Input Data Points}}$ → $\boxed{\text{Expected output}}$ → $\boxed{\text{Feedback signal}}$

Mapping the Tasks

Classification

Regression

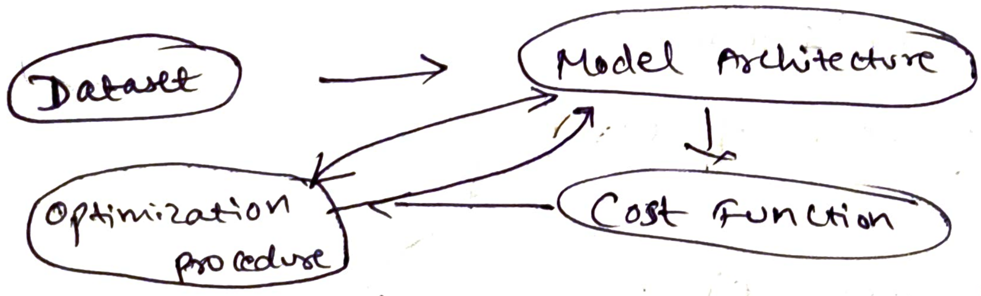
Transcription

Anomaly Detection

Categorization of learning Experience.

	Supervised Learning	Unsupervised Learning
Instructor Presence	Yes (Each Example is associated with a Provided label/target)	NO (Feature exist without a supervision signal)
primary Goal	Prediction & classification	Structure Discovery & Density Estimation
output Nature	Target Mappings	clusters & useful properties
Common Algorithms	Linear/Logistic Regression, SVMs	principal component Analysis (PLA)

Machine learning Recipe



Maximum likelihood Estimation

- Maximum likelihood Estimation is a parameter estimation technique that determines the values of model parameters by maximizing the likelihood function, which measures how well the model explains the observed data

Likelihood Function

$$L(\theta) = \prod_{i=1}^{n} P(x_i ; \theta)$$

where

$L(\theta)$: likelihood function

θ = Model parameters (weights)

$P(x_i ; \theta)$ = probability of observing sample x_i

why MLE used in Deep learning

Deep learning models must learn parameters that represent the training data.

MLE helps by:

→ Estimating optimal weights and biases
→ Improving prediction accuracy
→ providing a statistical basis for neural network

Negative Log-Likelihood (NLL)

$$NLL = -\log L(\theta)$$

ex:- correct class probability = 0.95

$NLL = -\log(0.95) = 0.051$

correct class probability = 0.050 (nearest value

Concept	Description	Example
Maximum likelihood	Estimates model parameters by maximizing likelihood of observed data	Neural Network training
likelihood function	Measure how likely the observed data is under a set of parameters	Probability of correct predictions

log-likelihood	Logarithm of the likelihood function for easier optimization	summing log probabilities
Negative log-likelihood (NLL)	Loss function obtained by negating the log-likelihood	classification loss

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Example 1: Coin Toss problem

suppose a coin is tossed 3 times. we assume the probability of getting a Head is $\theta = 0.8$ and Tail is $\theta = 0.2$. The observed results are: H(Head), H(Head), T(Tail). Find the likelihood of observing this data.

Probability of Head is $P(H) = 0.8$

Probability of Tail is $P(T) = 0.2$

likelihood function = $L(\theta) = P(H) \times P(H) \times P(T)$

$L(\theta) = 0.8 \times 0.8 \times 0.2$

$= 0.128$

Example 2: Student Pass prediction

suppose a machine learning model predicts the probability that a student will pass.

observed students:

Student	Probability Predicted
Student 1	0.90
Student 2	0.80
Student 3	0.95

All three students actually passed. Find the likelihood

likelihood

$$L(\theta) = 0.90 \times 0.80 \times 0.95$$
$$= 0.684$$

The model assigns a 68.4% likelihood to observed data

Building Machine learning Algorithm:

It is the systematic process of developing a Computational model that learns from historical data, identifies hidden patterns, and makes accurate predictions or decisions on new, unseen data.

Steps in Building a Machine learning Algorithm.

1. problem Definition

   Clearly defines the problem

2. Data Collection

   It require sufficient data for traning.

3. Data Preprocessing

   It improves data quality before model training.

4. Feature selection and Feature Engineering

   Features are input variables used by algorithm.
   selecting relevant features improves model performance.
   Feature Engineering creates new useful features from existing data.

5. splitting the Dataset.

   The dataset is divided into

   → Training set (70-80%)
   → Testing set (20-30%)

6. selecting a Machine learning Algorithm

Problem	Algorithm
classification	Decision Tree, SVM, Naive Bayes, Random Forest
Regression	Linear Regression, Decision Tree Regression
Clustering	K-Means, DBSCAN
Deep learning	CNN, RNN, LSTM

7. Training the Model.

During training, the algorithm learns patterns from the training data adjusting its internal parameters

8. Model evaluation

After training, the model is tested using unseen data.

True positive:

Model correctly predicts the positive class.

Ex: Actual: COVID positive predicted: COVID Positive

✓ Correct prediction

True Negative:

Model correctly predicts the negative class.

Ex: Actual: COVID Negative predicted: COVID Negative

✓ Correct prediction

False positive

Model incorrectly predicts positive class. The model predicts positive, but the actual is negative This is called Alaram,

Ex: - Real: No Disease Predicted - Disease

~~x~~ wrong prediction

False Negtive (FN):

Model incorrectly predicts the negative class.

The model predicts negative, but the actual class is positive. This is called ~~Miss~~ Missed Detection

Ex: Actual - Disease predicted - No Detection

~~x worng predictits~~

Confusion Matrix

It is a table used to evaluate the performance of a classification model by comparing the predicted labels with the actual labels.

Actual / predicted	positive	Negative
Positive	True positive (TP)	False Negative (FN)
Negative	False positive (FP)	True negtive (TN)

Accuracy

Accuracy measures the overall proportion of correctly classified instances

$$AP = \frac{TP + TN}{TP + TN + FP + FN}$$

Ex: TP = 50
TN = 40
FP = 5
FN = 5

$$\text{Accuracy} = \frac{50+40}{50+40+5+5} = \frac{90}{100} = 0.90$$

Accuracy = 90%.

Precision

Precision ~~makes~~ measures the ~~probabit~~ predicted cases ~~that~~ ar ~~actually~~.

$$\text{precision} = \frac{TP}{TP+FP}$$

$$\text{precision} = \frac{50}{50+10} = \frac{50}{60}$$

Recall

It measures the proportion of actual positive cases that are correctly identified

$$\text{Recall} = \frac{TP}{TP+FN}$$

Ex:

TP = 50
FN = 5

$$\text{Recall} = \frac{50}{55+5} = \frac{50}{55}$$

$= 0.9091$

F1 score

~~Harmonic pro~~ F1-score is harmonic mean of precision and Recall

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Permission} + \text{Recall}}$$

$$F1 = \frac{2 \times 0.80 \times 0.90}{0.80 + 0.90} = \frac{1.44}{1.70} = 0.847$$

Example

A school has developed a machine learning model to predict whether a student will Pass or Fail. After conducting the examination, the actual results are compared with the predicted results. The results of 10 students are shown below

Student	Actual Result	Predicted Result
S1	Pass	Pass
S2	Pass	Pass
S3	Pass	Fail
S4	Fail	Fail
S5	Fail	Pass
S6	Pass	Pass
S7	Fail	Fail
S8	Pass	Pass
S9	Fail	Fail
S10	Pass	Pass

Step 1: Identify TP, TN, FP and FN

We consider pass as the positive class

True positive (TP)

Actual = Pass

Predicted = Pass

Passed students:

S1

S2

S6

S8,

S10

Therefore,

TP = 5

True Negative (TN)

Actual = Fail

Predicted = Fail

Failed students

S4, S7, S9

Therefore,

TN = 3

False positive (FP)

Actual = ~~Pa~~ Fail

predicted = Pass

student

S5

Therefore

FP = 1

False Negative (FN)

Actual = Pass

Predicted = Fail

student

S3

Therefore

FN = 1

Step 2: Construct the confusion matrix

Actual / predicted	Pass	Fail
Pass	5 (TP)	1 (FN)
Fail	1 (FP)	3 (TN)

Step 3: Calculate Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Substitute the values

$$= \frac{5+3}{5+3+1+1} = \frac{8}{10} = 0.80$$

Accuray = 80%.

Step 4: Calculate precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

Substitute $= \frac{5}{5+1} = \frac{5}{6} = 0.8333$

~~Ac~~ Precision = 83.33%.

Step 5: Calculate Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

Substitute $= \frac{5}{5+1} = \frac{5}{6} = 0.8333$

Recall = 83.33%.

Step 6: Calculate F1-score

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= \frac{2 \times 0.8333 \times 0.8333}{0.8333 + 0.8333} = 0.8333$$

F1-Score = 83.33%.

Step 7: Calculate specificity

$$\text{specificity} = \frac{TN}{TN + FP} = \frac{3}{3+1} = \frac{3}{4} = 0.75$$

Specificity = 75%.

Final Result

Metric	Formula	value
TP	correct pass prediction	5
TN	Correct Fail prediction	3
FP	Incorrectly predicted Pass	1
FN	Incorrectly predicted Fail	1
Accuracy	(TP+TN)(TP+TN+FP+FN)	80%
Precision	TP(TP+FP)	83.33%

* A whether forecasting scheme containing 10 days ~~ext~~ periods forms the following model.

Day	Actual	Predict
Day 1	sunny	rainy
Day 2	rainy	rainy
Day 3	Sunny	rainy
Day 4	sunny	sunny
Day 5	rainy	sunny
Day 6	rainy	rainy
Day 7	rainy	sunny
Day 8	sunny	sunny

Step 1: Identify TP, TN, FP, FN

we consider sunny as positive class

True positive (TP)

Actual = sunny

Predicted = sunny

sunny forecasting

Day 4, Day 8

Therefore

TP = 2

True Negative

Actual = Rainy

Predicted = Rainy

Rainy Forecasting

Day 2, Day 6

Therefore TN = 2

False Negative (FN)

Actual = sunny

Predicted = Rainy

Forecasting

Day 1, Day 3

FN = 2

False positive (FP)

Actual = Rainy

Predicted = sunny

Forecasting

Day 5, Day 7

Therefore, FP = 2

Step 2: Construct the confusion matrix

Actual/predicted	Sunny ~~Pass~~	~~Fail~~ Rainy
sunny	2 (TP)	2 (FN)
Rainy	2 (FP)	2 (TN)

Step 3: Calculate Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{2+2}{2+2+2+2} = \frac{4}{8} = \frac{1}{2} = 0.5$$

Step 4: Calculate Precision

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{2}{2+2} = \frac{2}{4} = \frac{1}{2} = 0.5$$

Step 5: Calculate Recall

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{2}{2+2} = \frac{2}{4} = \frac{1}{2} = 0.5$$

Step 6: Calculate F1-score

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= \frac{2 \times 0.5 \times 0.5}{0.5 + 0.5} = 0.5$$

Step 7: Calculate specificity.

$$\text{specificity} = \frac{TN}{TN + FP} = \frac{2}{2+2} = \frac{2}{4} = \frac{1}{2} = 0.5$$

9. Model Deployment

Once the model ~~a~~ performs well, it is deployed for real-world use.

- It include platforms:
  - → web applications
  - → Mobile applications
  - → cloud platforms
  - → IoT devices

10. Model Monitoring and Maintenance.

The deployed model should be monitored continuously because data patterns may change over time.

Machine learning Mandate

The machine learning Mandate refers to the fundamental principle that computers should learn pattern, relationships, and decision-making rules automatically from data rather then being explicitly programmed with handcrafted rules

Underfitting and overfitting in Machine learning

During model training, two common problems may occur : underfitting and overfitting

underfitting

It occurs when a machine learning model is

-> Too simple
-> High training and testing error can happen
-> poor prediction accuracy

Overfitting

overfitting occurs when a machine learning model learns not only the actual patterns but also the noise and random fluctuations present in the training data. As a result, the model performs well on training data but poorly on new, unseen data

Ex: Image Classification

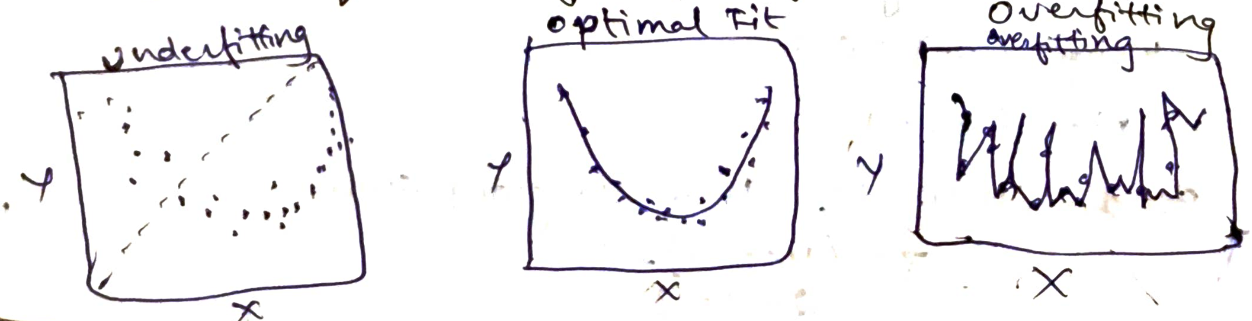
Suppose a model is trained to recognize cats.

Training Images: 500 cat images

The model memorizes:

-> Background color
-> camera angle
-> lighting

The consequences of capacity: Underfitting and Overfitting



No Free Lunch (NFL) Theorem

→ NFL Theorem which states that no single machine learning algorithm is best for every problem or dataset

→ NFL proposed by David

Forward propagation:

Process of neural network where input data is passed from the input layers.

Steps:

- Input data is provided to the network.
- Each neuron calculates the weighted sum of inputs
- ~~Eac~~ A bias is added
- An activation function is applied
- The output is passed to the next layer

$$z = \sum_{i=1}^{n} w_i x_i + b$$

Ex: Student Pass Prediction

Suppose a neural network predicts whether a student will pass or fail based on study hours

Given

Input (study Hours) : 4

weight (w) = 0.5

Bias (b) = 1

Step 1: Calculate the weighted sum

$z = (4 \times 0.5) + 1$

step 2: Apply the Activation function ...

Assume we use the ReLU (Rectified linear unit) activation function

$$z = 3$$

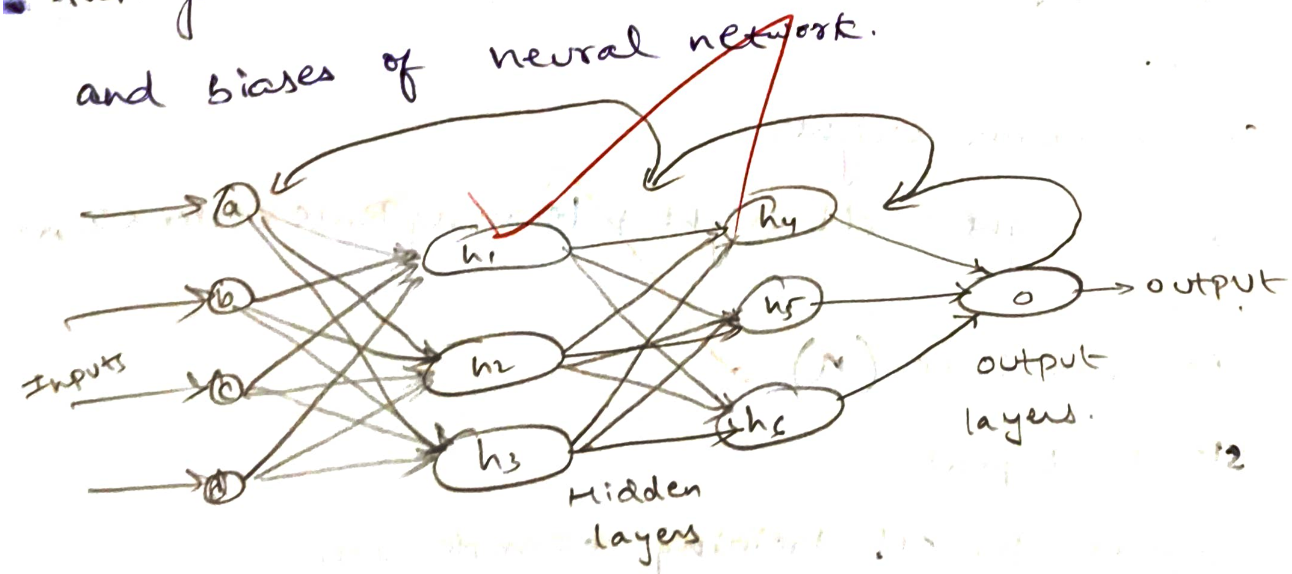
$$ReLU(3) = 3$$

Output:

The network produces an output of 3. Based on predefined decision rule, this value may indicate that the student is likely to pass.

Backpropagation

It is a learning algorithm that calculates the error at the output layer and propagates it backward through the hidden layers to update the weights and biases of neural network.



steps:

step 1: Initialize weights

Assign small ~~roor~~ random values to all weights

Ex: $w1 = 0.2$ $W2 = -0.3$ $W3 = 0.5$

step 2: Feed Input

Provide training data to the network.

example : Input = (2,3)

Step 3: Forward pass

Calculate neuron outputs.

Net Input = Σ (Input x weight).

output = Activation (Net Input)

Step 4: calculate Error

Error is calculated using

Error = Actual output - predicted output.

Example:

Actual output = 1

Predicted output = 0.82

Error = 1 - 0.82

= 0.18

Step 5: Compute Gradient.

Determine how much each weight contributed to the error.

Step 6: Update weights

New weight = old weight + learning Rate x Error x Input

where

Learning Rate (η)

Step 7: Repeat

Repeat for all training examples until

Error is minimum

Maximum epochs reached

Q: Calculate the output of a single Neuron

Given:

- Input x = 2
- Weight w = 0.4
- Bias b = 0.2
- Activation function = sigmoid

Find the output

step 1 : Net Input

Net = xw+b

$(2)(0.4)+0.2 = 0.8+0.2 = 1.0$

step 2 : sigmoid

$$\text{output} = \frac{1}{1+e^{-1}} = \frac{1}{1+0.3679} = 0.731$$

Q : Calculate the Error

Given

Actual output = 1

predicted output = 0.75

Find the error

Error = Target - output

$= 1-0.75 = 0.25$

Q : Find out Mean squared Error (MSE)

Given: Target outputs 1, 0, 1

predicted outputs 0.8, 0.3, 0.9 Find MSE

$$MSE = \frac{1}{3}\left[(1-0.8)^2+(0-0.3)^2+(1-0.9)^2\right]$$

$$= \frac{1}{3}(0.04+0.09+0.01) = \frac{0.14}{3} = 0.046$$

Q. Find out weight update

Given

old weight = 0.50

learning Rate = 0.20

Error = 0.40

Input = 1

Find the new weight

Newweight = Oldweight + η + Error + Input
$= 0.5 + (0.2)(0.4)(1)$
$= 0.5 + 0.08 = 0.58$

Linear Algebra

1. Scalar

→ Scalar is a single numerical value that has only magnitude and no direction

→ Represented by lowercase letter

2. Vector

→ A vector is an ordered collection of numbers arranged in a single row or column.

Types of vectors.

1. Vector addition.

$A = [2 \ 4 \ 6]$ $B = [1 \ 3 \ 5]$

$A + B = [3 \ 7 \ 11]$

2. Vector subtraction

$A = [2 \ 4 \ 6]$ $B = [1 \ 3 \ 5]$

$A - B : [1 \ 1 \ 1]$

3. Scalar Multiplication

$A = [2 \ 4 \ 6]$

Multiply by 3 $3A = [6 \ 12 \ 18]$

Matrix

A matrix is a rectangular arrangement of numbers organized into rows and columns.

Matrix Addition

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 3 & 5 \\ 7 & 9 \end{bmatrix}$$

Matrix subtraction

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$A - B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Matrix Multiplication

It is the process of multiplying row of first mat with column of ~~first~~ second matrix

Matrix $A = m \times n$, Matrix $B = n \times p$

Result $m \times p$

Scalar Multiplication.

A machine learning model has weight of 4.

Vector Addition

Two students have feature vector:

$A = [2 \ 4 \ 6]$

$B = [1 \ 3 \ 5]$ $\quad A + B$

$A + B \ [2 + 1 \ \ 4 + 3 \ \ 6 + 5]$

$= [3 \ \ 7 \ \ 11]$

Neural Network forward propogation

$y = xw + b$ [2

Input x : [2   3]

weight $w = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$

$b = 2$

$y = xw + b = [2 \times 4] + [3 \times 5] + 2 = 8 + 15 + 2 = 25$

Hidden performance [x =28%

weights $w_1 = 2$

$w_2 = 0.1$

Bias : 3

$y = (5 \times 2) + (90 \times 0.1) + 3$

$= 10 + 9 + 3 = 22$

Ex:

Predicting salary using linear model

salary = weight $\times$ Experience + bias

$= 8 \times 6 + 5$

$8 \times 11 = 88$

Stochastic Gradient Descent (SGD)

It is a variant of Gradient Descent in which the model updates its weights after processing each individual training sample, rather than waiting for the entire dataset.

working of SGD

x_1	10
x_2	15
x_3	20

Training sequence :

Input x_1 → update weight

Input x_2 → update weight

Input x_3 → update weights

Mini Batch Gradient Descent

Instead of using one sample or the entire dataset, Mini-Batch Gradient Descent uses a small batch of samples.

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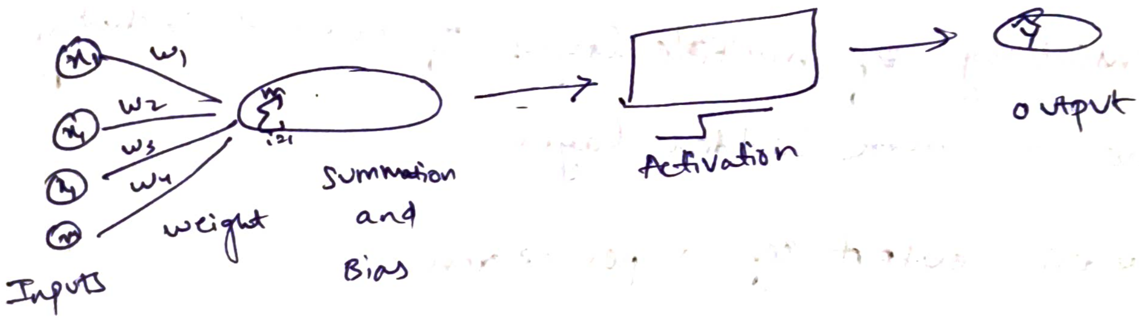
Perceptrons

A perceptron recieves one or more input values, multiplies each input by a corresponding weight, adds a bias, and applies an activation function to produce an output.



structure of a perceptron.

→ Input layer

→ weights

→ Bias

→ Summation unit

→ Activation Function

Steps of the percepton Algorithm.

step 1 : Initialize weights

step 2 : provide Training Data

step 3 : ~~[illegible]~~ Compare the New Input

$$z = \sum_{i=1}^{n} w_i x_i + b$$

step 4: Apply Activation ~~hidden~~ Function

step 5: Compare with Target ~~testy~~ output.

step 6: Update weights

$$w_{new} = w_{old} + \eta(t-y)x$$

$$b_{new} = b_{old} + \eta(t-y)$$

step 7: Repeat

Multilayer Perceptrons (MLP)

It is an advanced type of neural network consisting of multiple layers of neurons, including one or more hidden layers.

Calculate output of a perceptron

Inputs $x_1 = 2$, $x_2 = 3$

Weights $w_1 = 0.5$, $w_2 = 0.4$

$b = -1$

Output $= 1$ if net ≥ 0

output $= 0$ otherwise

(1) compute Net Input

$$Net = x_1 w_1 + x_2 w_2 + b$$

~~2(0.5) + 3(0.4)~~

$$2(0.5) + (3)(0.4) - 1 = 1 + 1.2 - 1 = 1.2$$

Since Net ≥ 0

→ Update the weight using perceptron learning Rule

Given

learning Rate = 0.2

Target = 1 (predicted)

output (Actual) = 0

Input = 3 , old weight = 0.4

$$w_{new} = w_{old} + \eta(\text{Target} - \text{output})x$$

$$= 0.4 + 0.2(1-0)(3)$$

$$= 0.4 + 0.6$$

$$= 1$$

→ update Bias

Given

old Bias = 0.5 , learning Rate = 0.1

Target (predicted) = 0 , output (Actual) = 1

$$b_{new} = b + \eta(\text{Target} - \text{output})$$

$$= 0.5 + 0.1(0-1)$$

$$= 0.5 - 0.1$$

$$= 0.4$$

Curse of dimensionality

The curse of dimensionality refers to the challenge that arise when working with high-dimensional data, where increasing the number of features leads to, reduced model performance and higher computation costs.

* It becomes more difficult for the model to identify meaningful patterns

How to overcome the curse of Dimensionality

→ Feature selection

- Choose only the most relevant features.

→ Use Appropriate Algorithms

- Some algorithms are better suited for high-dimensional data

→ Dimensionality Reduction

- Reduce the number of features while retaining most of the important information

Probability & Statistics for Machine Learning

1. probability:

Probability is the measure of how likely an event is to occur. The probability of any event lies between 0 and 1.

$$P(E) = \frac{\text{No. of favourable outcomes}}{\text{No. of possible outcomes}}$$

2. Probability Rules

(a) Addition Rule.

$$P(A \cup B) = P(A) + P(B)$$

(b) Multiplication Rule

$$P(A \cap B) = P(A) \times P(B)$$

(c) complement Rule

$$P(A') = 1 - P(A)$$

~~Ex:~~

3. Random Variable

A Random variable is a variable whose value depends on the outcome of a random experiment

(a) Discrete Random variable

Ex :- 0, 1, 2, 3 ....

(b) continuous Random Variable

Ex: 30.5°C, 31.2°C, 32.7°C

4. probability distribution

→ Bernoulli Distribution

→ Binomial Distribution

→ Normal Distribution

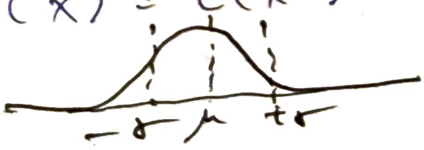
→ Poisson Distribution

5. Mean

$$\text{Mean} = \frac{\sum x}{n}$$

6. Variance :- Measures how far the data values are spread from mean

$$\text{var}(x) = E(x^2) - [E(x)]^2$$



7. Standard Deviation

~~It is~~

$\sigma = \sqrt{\text{variance}}$ → feature selection and measuring data spread

Ex:

vaiance = 2.67

standard deviation = $\sqrt{2.67}$

= 1.63

Real-Time Machine leaning example.

student	Predicted Marks
A	70
B	75
C	80
D	85
E	90

$$\text{Mean} = \frac{70+75+80+85+90}{5} = 80$$

variance =

$$\sigma^2 = N \sum (X-\mu)2$$

Marks (x)	Mean (μ)	(x-\mu)	((x-\mu)^2)
70	80	-10	100
75	80	-5	25
80	80	0	0
85	80	5	25
90	80	10	100

~~standard~~

sum of squared Deviations

100+25+0+25+100 = 250

Divide by No. of observations

$$\sigma^2 = \frac{250}{50}$$

$$= 50$$

$\boxed{\text{Variance} = 50}$

Find the standard Deviation

$\sigma = \text{variance}$

$\sigma = \sqrt{50}$

$= 7.07$

standard deviation ≈ 7.07

Supervised learning Algorithm

→ It is a type of Machine learning in which a model learns from labeled training data

Types of Supervised learning

a) Regression

→ used to predict continuous numerical output based on one or more input variables

→ predicts a continuous numerical value

Ex:

→ House price prediction

→ Temperature prediction

Ex:-

Predicting house ~~pred~~ prices based on the area of the house

House Area → House price

linear regression eq^n - $y = mx + c$

Predicted output, slope, Input, Intercept

b) Classification

→ It is supervised learning technique used to predict a categorical output or class label.

Ex:

- Spam or Not spam
- pass or Fail

Types of classification

(i) Binary classification

(ii) Multi-class classification

Decision Tree

→ It is a supervised learning algorithm that uses a tree-like structure to make predictions

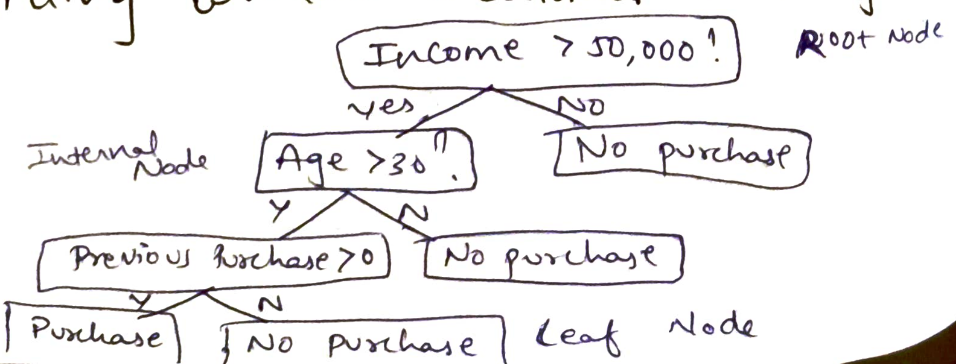
→ It can be used for both classification and regression

The tree consists of:

- Root Node
- Decision / Internal Node
- Branch
- Leaf Node

Ex:-

Predicting whether a customer will buy a prod



K-Nearest Neighbors (K-NN)

→ It is a supervised learning algorithm that predicts the class of a new data point based on the classes of its k nearest data points

→ It is based on the idea:

similar data points are likely to belong to the same class.

K-NN commonly uses Euclidean Distance.

Euclidean distance formula

$A(x_1, y_1) \qquad B(x_2, y_2)$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

K-NN problem

class A : A (1,1)

class B : B (2,2)

class C : C (6,6)

P (2,1)

let k = 3

Find predicted class

Ans

P (2,1) A (1,1)

$$d = \sqrt{(2-1)^2 + (1-1)^2} = A = \sqrt{1} = 1$$

P (2,1) B (6,6)

$$d = \sqrt{(2-6)^2 + (1-6)^2} = \sqrt{16+25} = \sqrt{41} \approx 6.4$$

P (2,1) C (6,6)

$$d = \sqrt{(2-6)^2 + (1-6)^2} = \sqrt{16+25} \approx 6.4$$

Voting

Votes

A = 1

B = 1

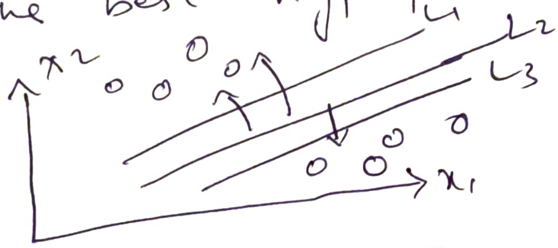
C = 6.4

Predicted class = A and B

Support Vector Machine (SVM)

→ It is a supervised learning algorithm mainly used for classification, although it can also be used for regression.

→ Find the best hyperplane



* The data points closest to the decision boundary are called support vectors.

Key Terms

1. Hyperplane

The boundary that separates different classes.

$$w_1 x_1 + w_2 x_2 + b = 0$$

2. Margin

The distance b/w decision boundary and the nearest data points from each class.

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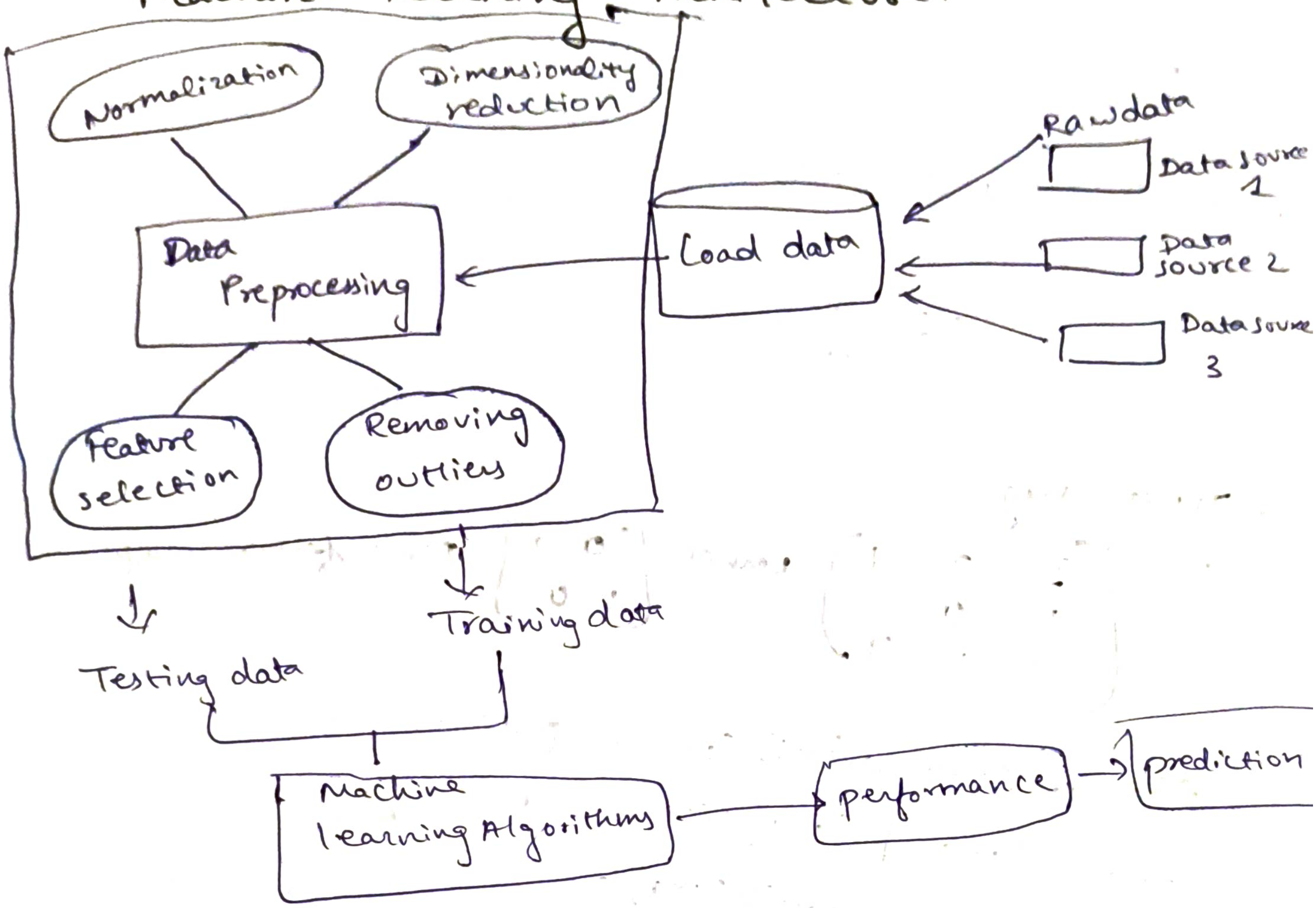
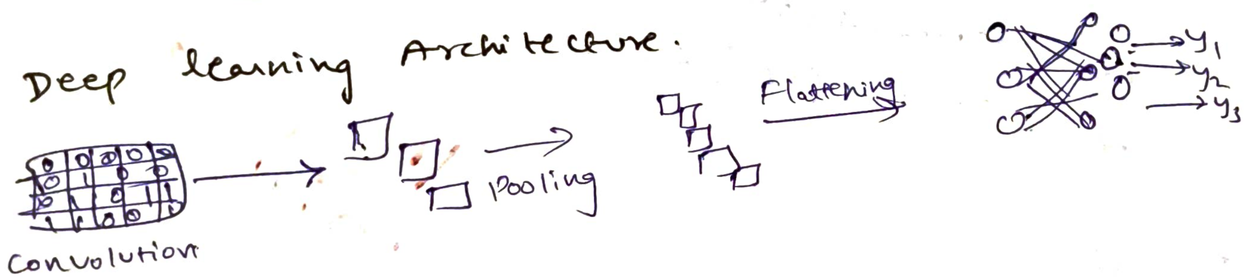
Machine learning Architecture.



Deep learning Architecture.



Feature Map

A feature Map is the output generated by applying a convolutional filter to an input image or a previous layer's output within a CNN

Ex:- Input Image 3x3

$$I = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

kernel 2x2

$$K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Find Feature map

First position

$$\begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$$

Multiplication $(1\times1)+(2\times0)+(4\times0)+(5\times1)$

$= 1+0+0+5 = 6$

Move Right $\begin{bmatrix} 2 & 3 \\ 5 & 6 \end{bmatrix}$

$= (2\times1)+(3\times0)+(5\times0)+(6\times1)$

$= 2+6 = 8$

Move Down

$$\begin{bmatrix} 4 & 5 \\ 7 & 8 \end{bmatrix}$$

$= (4\times1)+(5\times0)+(7\times0)+(8\times1)$

$= 4+8 = 12$

Last position

$$\begin{bmatrix} 5 & 6 \\ 8 & 9 \end{bmatrix}$$

$: (5\times1)+(6\times0)+(8\times0)+(9\times1)$

$= 5+9 = 14$

$$= \begin{bmatrix} 6 & 8 \\ 12 & 14 \end{bmatrix}$$

* Input Image

$$\begin{bmatrix} 3 & 2 & 1 \\ 1 & 0 & 2 \\ 2 & 1 & 3 \end{bmatrix}$$ kernel $$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$ Find feature Map.

$(3\times0)+(2\times1)+(1\times1)+(0\times0) = 0+2+1 = 3$

$(2\times0)+(1\times1)+(0\times1)+(2\times0) = 1$

$(2\times0)+(1\times1)+(2\times1)+(1\times0) = 2$

$(1\times0)+(1\times0)+(2\times1)+(1\times0)$

$(1\times0)+(1\times0)+(2\times1)+(3\times0) = 3$

$(0\times0)+(2\times1)+(1\times1)+(3\times0)$

Ans $\begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}$

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Unsupervised learning:- It is a type of Machine learning in which the algorithm learns patterns and structures from unlabeled data.

1. K-Means clustering

→ It is an unsupervised learning algorithm that divides a dataset into K-clusters based on the similarity b/w data points.

→ Each cluster is represented by centroid, which is the mean of data points belonging to cluster.

K → No. of clusters

centroid → center of cluster

cluster → Group of similar data points

Euclidean Distance

$A(x_1, y_1)$ $B(x_2, y_2)$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Problem: K-Means clustering

Consider the following four data points:

$A(1,1)$, $B(2,2)$ $C(8,8)$ $D(9,9)$

let : $K=2$

Assume the initial centroids are : $C_1 = (1,1)$ $C_2 = (8,8)$

Assign each point to the nearest cluster.

Step 1: Calculate Distances for $A(1,1)$

Distance from $C_1 = (1,1)$: $A(1,1)$

$d(A, C_1) = \sqrt{(1-1)^2 + (1-1)^2}$

Cluster 1 = 0

Distance from $C_2 = (8,8)$: $A(1,1)$

$d(A, C_2) = \sqrt{(8-1)^2 + (8-1)^2} = \sqrt{49+49} = \sqrt{98}$

Cluster 2 = $\sqrt{98} \approx 9.90$

Therefore, A belongs to cluster 1

Step 2: Calculate Distances from $B(2,2)$

Distance from $C_1 = (1,1)$ $B(2,2)$

$d(B, C_1) = \sqrt{(2-1)^2 + (2-1)^2} = \sqrt{1^2+1^2} = \sqrt{2}$

Calculate Distances from $B(2,2)$

Distance from $C_2 = (8,8)$ $(2,2)$

$d(B, C_2) = \sqrt{(8-2)^2 + (8-2)^2} = \sqrt{72}$

Therefore, B belongs to cluster 1

Step 3: Calculate Distance for $C(8,8)$

Distance from $C_1(1,1)$ $(8,8)$

$d(C,C_1) = \sqrt{(8-1)^2+(8-1)^2} = \sqrt{7^2+7^2} = \sqrt{49+49} = 9.9$

Distance from $C_2 = (8,8)$ $C(8,8)$

$d(C,C_2) = 0$

Cluster 2 : 0

Therefore, C belongs to cluster 2 (small value)

Step 4: calculate Distance for $D(9,9)$

Distance from $C_1(1,1)$ $D(9,9)$

$d(C_1,D) = \sqrt{(9-1)^2+(9-1)^2} = \sqrt{8^2+8^2} = \sqrt{128} \approx 1.3$

Distance from $C_2(8,8)$ $D(9,9)$

$d(C_2,D) = \sqrt{(9-8)^2+(9-8)^2} = \sqrt{1^2+1^2} = \sqrt{2}$

There, D belongs to C_2

Step 5: Recalculate Centroids

For cluster 1 : $A(1,1)$ $B(2,2)$

$C_1 = \left(\frac{1+2}{2}, \frac{1+2}{2}\right) \Rightarrow C_1 = (1.5, 1.5)$

For cluster 2 : $C(8,8)$, $D(9,9)$

$C_2 = \left(\frac{8+9}{2}, \frac{8+9}{2}\right) \Rightarrow C_2 = (8.5, 8.5)$

9. Hierarchical clusteiing

It is an unsupervised learning technique that creates a hierarchy of clusters.

Problem: Hierarchical clustering

Consider four points:

$A=1$, $B=2$, $C=8$, $D=9$ use hierarchical clustering to form two clusters.

Step 1: Calculate Distances

$d(A, B) = |1-2| = 1$

$d(A, C) = |1-8| = 7$

$d(A, D) = |1-9| = 8$

$d(B, C) = |2-8| = 6$

$d(B, D) = |2-9| = 7$

$d(C, D) = |8-9| = 1$

Step 2: Find the closest points

the smallest distance is

$d(A, B) = 1$ $\quad$ $d(C, D) = 1$

Therefore merge: $\{A, B\}$ and $\{C, D\}$

There are two types of Hierarchial Clustering

1. Agglomerative Clustering

It follows a bottom-up approach

Each data point = one cluster

2. Divisive Clustering

It follows a top-down approach

All data points = one cluster.

3. Dimensionality Reduction

It is the process of reducing the number of features or dimensions in a dataset while preserving as much important information as possible.

4. Principal Component Analysis (PCA)

It is a dimensionality reduction technique that transforms a large number of correlated features into a smaller number of uncorrelated variables called Principal Components

PCA selection

Principal Component	Explained variance
PC1	60%
PC2	25%
PC3	10%
PC4	5%

Find the number of principal components required to retain at least 95% of the variance.

Using PC1 : 60%.

Using PC1 + PC2 : 60 + 25 = 85%.

Using PC1 + PC2 + PC3 : 60 + 25 + 10 = 95%.

They retain : 95% variance.

Represent learning :- It is the process of automatically discovering and extracting meaningful features from raw data that help a Deep learning model make accurate predictions.

Raw Data
↓
Representation learning
↓
useful feature Representation
↓
Machine learning Model → Final Pred.

Types of Representations learned
→ low-level features
→ Mid-level features
→ High-level features

Neural Networks Design

→ It is a computational model consisting of interconnected neurons organized into layers.

→ It learns by adjusting the connection weights between neurons based on training data, enabling to recognize patterns

Biological Inspiration

Human Brain	Artificial Neural Network
Neuron	Artificial Neuron
Dendrites	Input Nodes
Cell Body	processing unit
Axon	Output Node
synapse	weight

Three important Concepts in Neural network design:

→ width & Depth of Neural Networks

→ Model capacity

→ Network Architecture.

Example problem.

Consider the following network:

$4 \to 8 \to 6 \to 2$

Input layer = 4 neurons

Hidden layer 1 = 8 neurons

Hidden layer 2 = 6 neurons

Output layer = 2 neurons

∴ width of Hidden layer 1 = 8

width of Hidden layer 2 = 6

The maximum hidden-layer width is: 8

Recurrent Neural Network (RNN)

- processes sequential data
- Maintains memory of previous inputs
- used in language translation and speed recognition.
- Reccurent Neural Network is a type of artificial ~~intel~~ neural network designed to Process sequential data.

Deep Neural Network.

- contains multiple hidden layers.
- can learn highly complex representations.
- used in deep learning applications.

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* A Restricted Boltzmann machine (RBM) is a type of generative neural network used to learn patterns and representations from data.
* RBMs are unsupervised learning models, meaning they learns from data without requiring labeled target outputs.

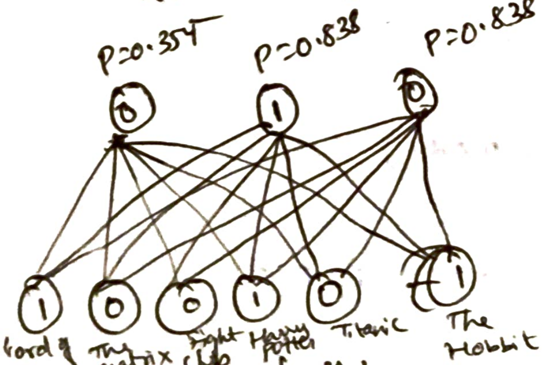
An RBM consists of two layers:

1. visible layer
2. Hidden layer

* The two layers are connected, but there are no connections between neurons within the same layer.

$\boxed{\text{Visible layer} \longleftrightarrow \text{Hidden layer.}}$

Hidden layer



Visible layer.

Visible Nodes - Bottom layer

The bottom layer contain six movies:

- Lord of the Rings → 1
- The Matrix → 0
- Fight club → 0
- Harry potter → 1
- Titanic → 0
- The Hobbit → -1

1 → User likes the movie

0 → User has not watched

-1 → User dislikes the movie.

Probability of Hidden units in RBM:

$$P(h_j = 1 | v) = \sigma(b_j + \sum_i w_{ij} v_i)$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$

Example: ① $v_1 = 1$, $v_2 = 0$

weights are:
$w_1 = 0.5$, $w_2 = 0.3$

Hidden bias: $b = 0.1$. Calculate the activation probability of hidden unit.

step 1: $z = b + w_1 v_1 + w_2 v_2$

Substitute: $z = 0.1 + (0.5)(1) + (0.3)(0)$

$z = 0.1 + 0.5 + 0 = 0.6$

step 2: Apply sigmoid

$P(h = 1/v) = \frac{1}{1+e^{-0.6}}$ we know, $e^{-0.6} \approx 0.5488$

Therefore:
$P(h=1/v) = \frac{1}{1+0.5488} = \frac{1}{1.5488}$

$P(h=1/v) \approx 0.646$

∴ The probability that hidden unit is activated approx 64.6%.

② $\eta = 0.1$, $v_i = 1$, $h_j = 1$, $v_i' = 0$, $h_j' = 1$

calculate weight update.

Sol Formula: $\Delta w_{ij} = \eta(v_i h_j - v_i' h_j')$

$= 0.1[(1)(1) - (0)(1)]$

$= 0.1(1-0) \Rightarrow \boxed{\Delta w_{ij} = 0.1}$

initial weight is $w_{ij} = 0.5$ then $w_{ij}^{new} = w_{ij} + \Delta w_{ij}$

$= 0.5 + 0.1 = 0.6$

③ $v_1 = 1$, $v_2 = 0$, $w_1 = 0.5$, $w_2 = 0.3$, $b = 0.1$

Calculate the probability that the hidden unit is active

Sol $$P(h=1|v) = \sigma(b + w_1 v_1 + w_2 v_2)$$

where:
$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$z = 0.1 + (0.5)(1) + (0.3)(0)$$
$$z = 0.6$$

Therefore: $$P(h=1|v) = \frac{1}{1+e^{-0.6}} \approx 0.646$$

$$\boxed{P(h=1|v) \approx 0.646}$$

④ $v_1 = 1$, $v_2 = 1$, $w_1 = 0.4$, $w_2 = 0.6$, $b = -0.2$

Calculate the hidden unit activation probability.

Sol $$z = b + w_1 v_1 + w_2 v_2$$
$$z = -0.2 + (0.4)(1) + (0.6)(1) = -0.2 + 0.4 + 0.6$$
$$\boxed{z = 0.8}$$

Apply sigmoid: $$P(h=1|v) = \frac{1}{1+e^{-0.8}} \approx 0.690$$

The hidden unit has approximately a 69% probability of activation

⑤ $v_1 = 1$, $v_2 = 0$, $v_3 = -1$, $w_1 = 0.5$, $w_2 = 0.2$, $w_3 = -0.4$, $b = 0.1$

Calculate hidden unit activation probability.

Sol $$z = b + w_1 v_1 + w_2 v_2 + w_3 v_3$$
$$z = 0.1 + (0.5)(1) + (0.2)(0) + (-0.4)(-1)$$
$$z = 0.1 + 0.5 + 0 + 0.4 \Rightarrow z = 1.0$$

Apply sigmoid
$$P(h=1|v) = \frac{1}{1+e^{-1}} \approx 0.731$$

∴ The hidden unit has approx 73.1% Probability of activation